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Street Vendor Digitalization Agent – using IBM Watsonx with IBM BOB

Create a new AI Agent – Street Vendor Digitalization Agent

Create a new AI Agent – a Street Vendor Digitalization Agent that helps local hawkers and micro-entrepreneurs establish a digital presence by generating business profiles, providing UPI payment setup guidance, recommending local SEO strategies, retrieving MSME schemes, and offering customer engagement suggestions using Retrieval-Augmented Generation (RAG).

Description

The Street Vendor Digitalization Agent is an AI-powered assistant built using IBM Watsonx.ai and IBM Granite models. It enables street vendors to grow their businesses digitally by creating professional business profiles, guiding them through UPI QR code setup, suggesting Google Business Profile registration, improving local search visibility, and generating promotional content.

Using Retrieval-Augmented Generation (RAG), the agent retrieves real-time information such as MSME schemes, government policies, digital onboarding procedures, geolocation-based business insights, and customer behavior trends from trusted government portals and commerce platforms.

A vendor can simply enter a prompt such as "I sell fruits in Pune Camp area." The AI agent will generate a business profile, UPI QR code setup guide, Google Business Profile recommendations, local SEO strategies, MSME scheme suggestions, pricing recommendations, promotional content, and customer engagement tips.

This AI-powered solution helps informal businesses increase their digital visibility, improve customer reach, and access government support programs.

Tools & Technologies Used

IBM watsonx.ai

IBM Granite Models

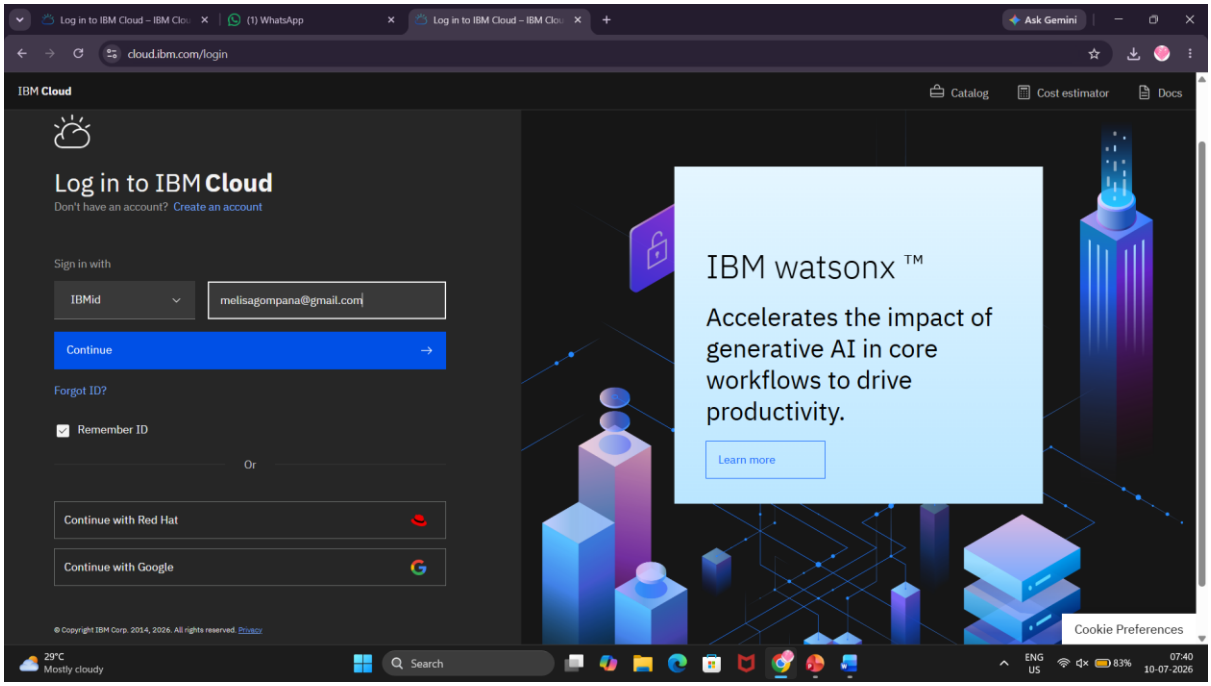
IBM Cloud Lite

IBM BOB

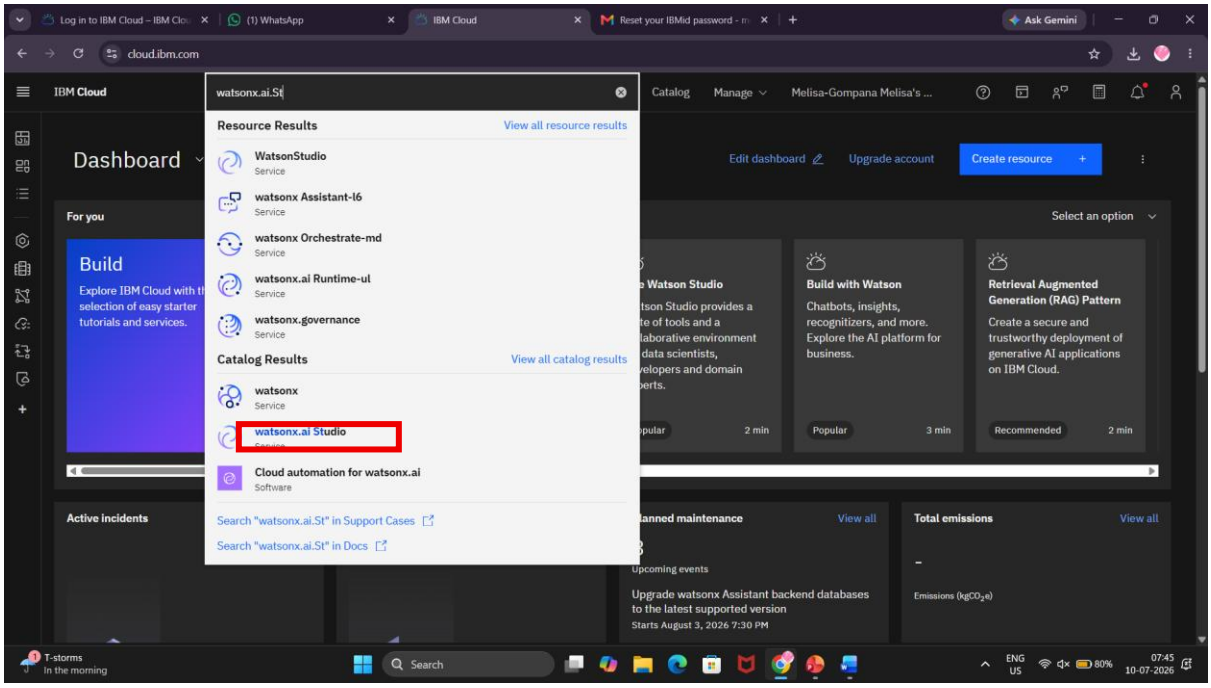
Python Flask

Retrieval-Augmented Generation (RAG)

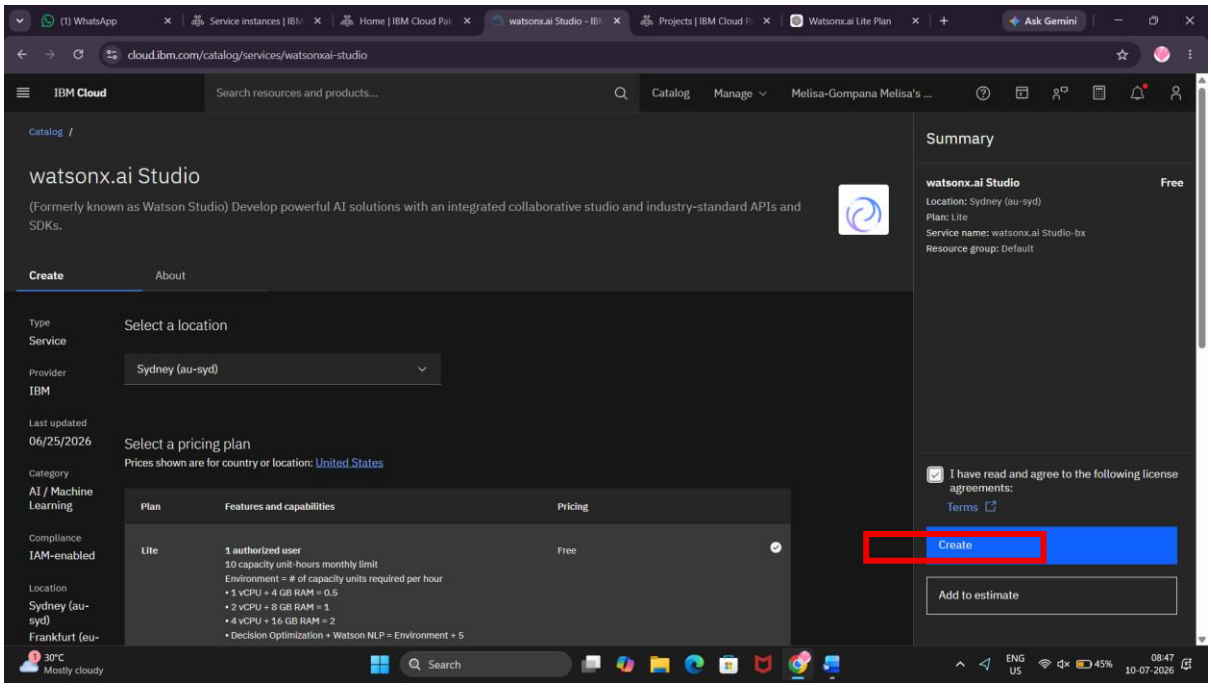
Step-1 login with your credential on IBM cloud.



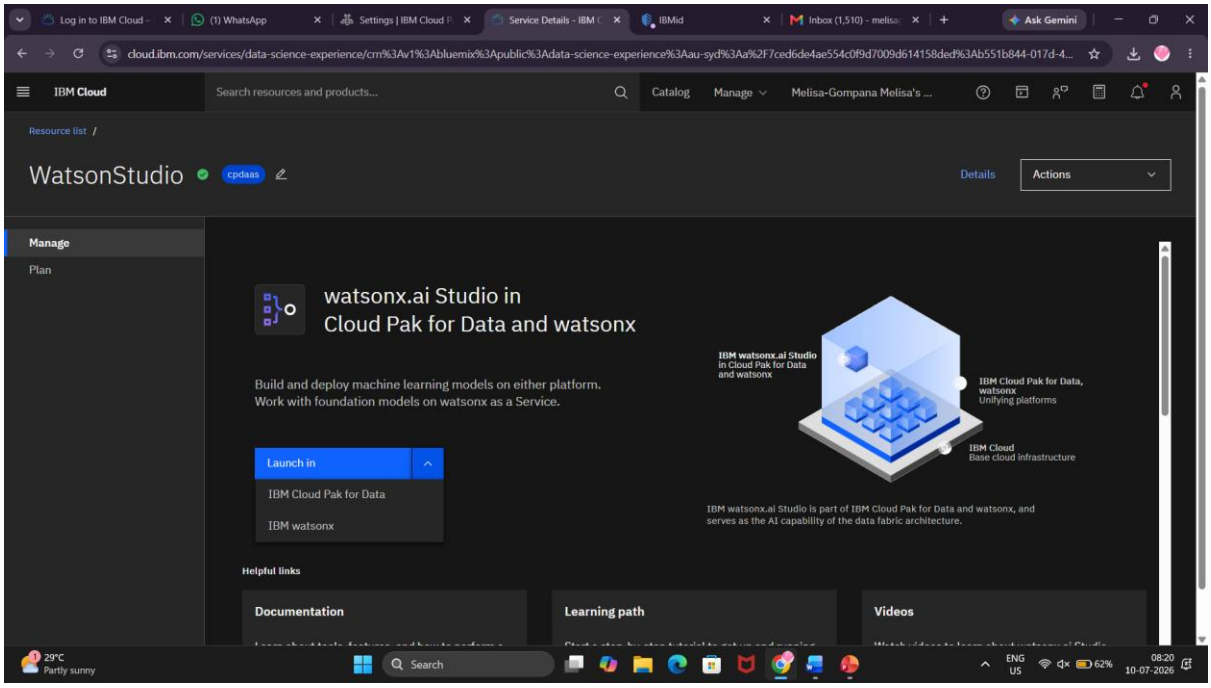
Step-2 go to search bar and search **watsonx.ai.studio** .



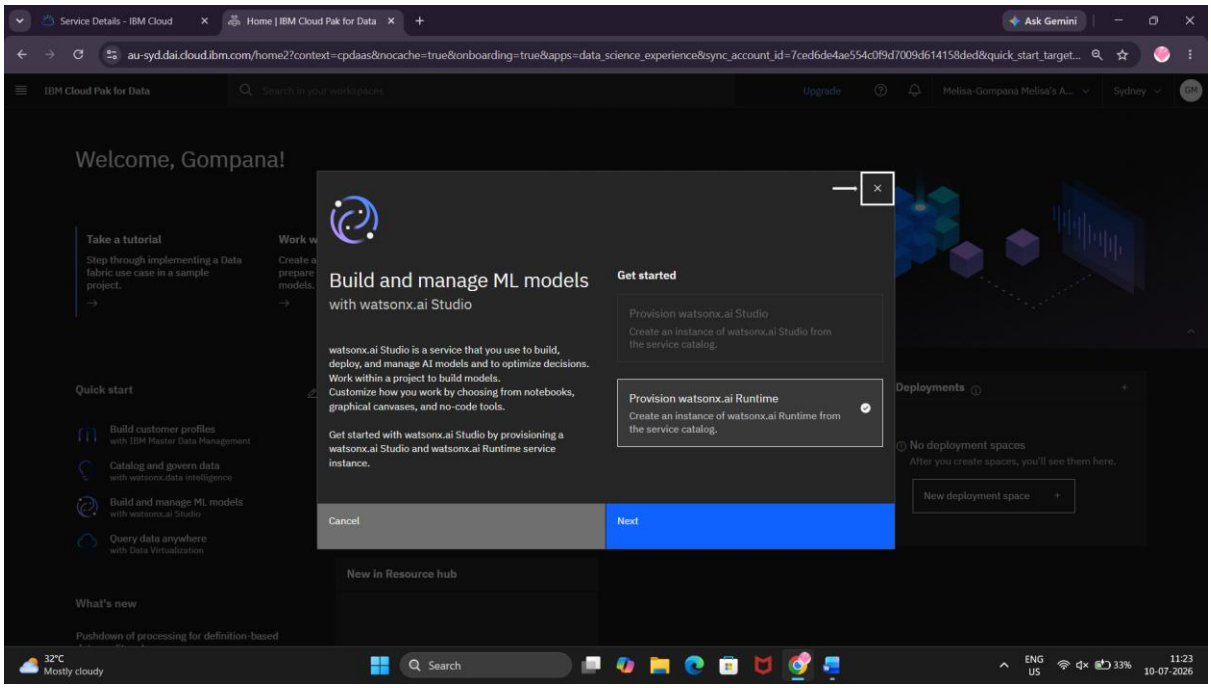
Step-3 create watsonx service select location as Sydney go with free plan and mark checkbox and create.



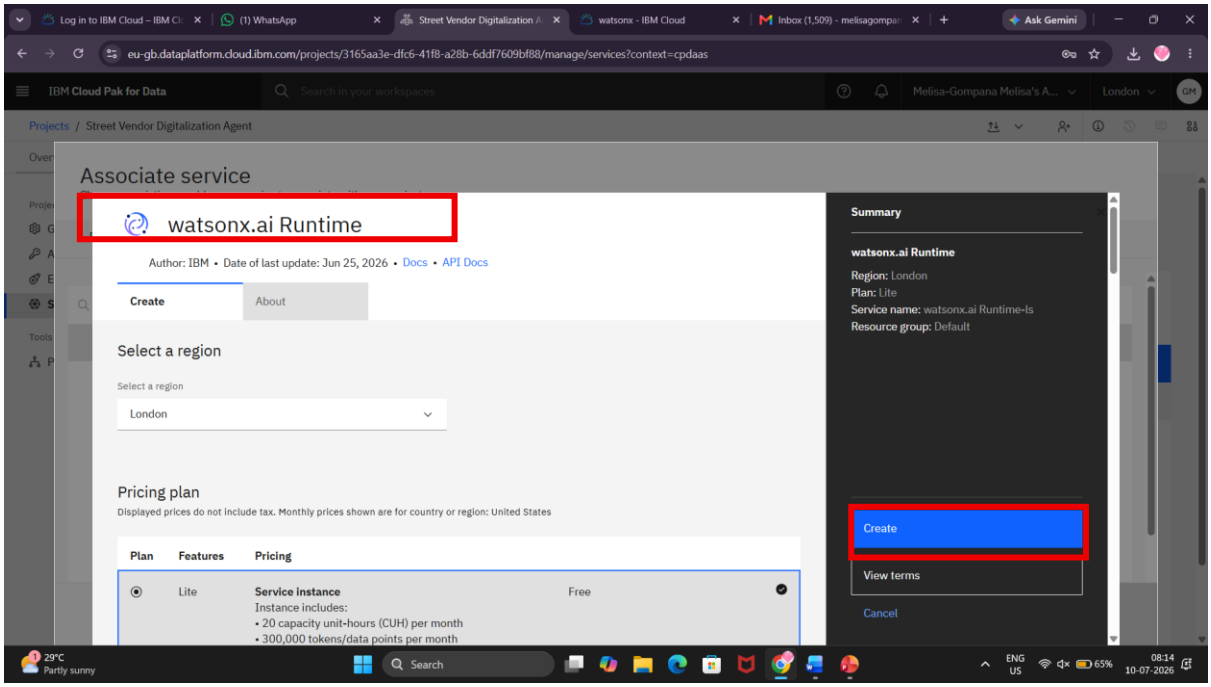
Step-4 click on **launch in** button



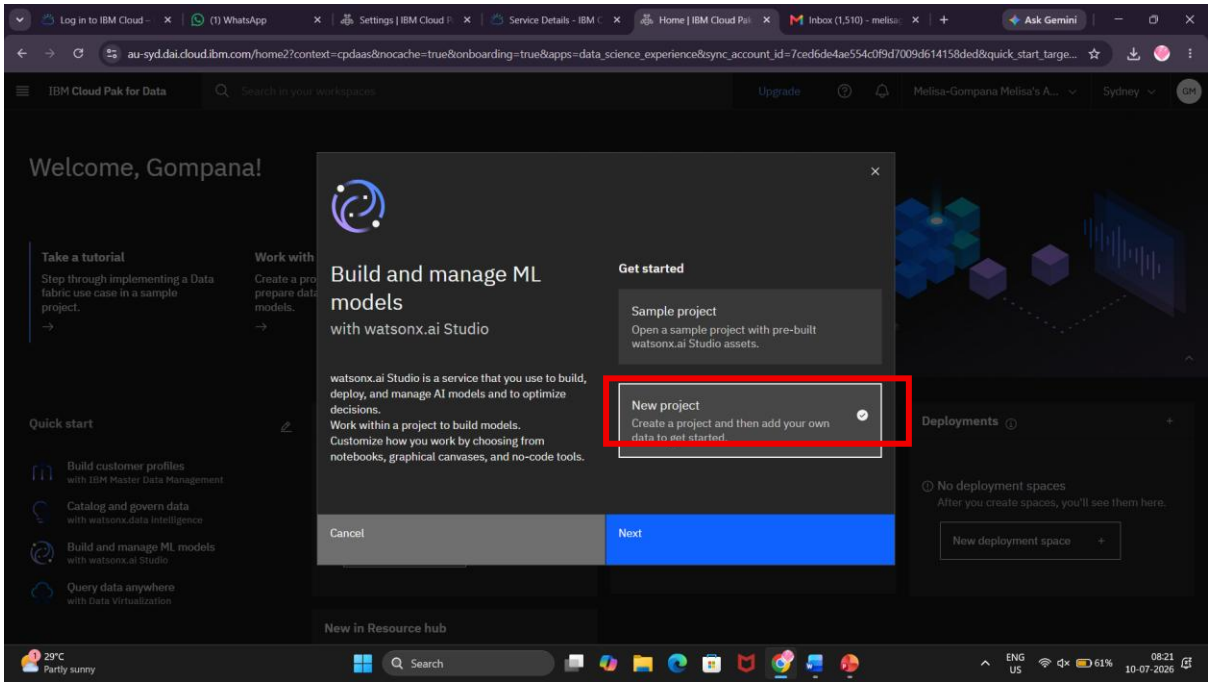
Step-5 Select **watsonx runtime** and click on next.



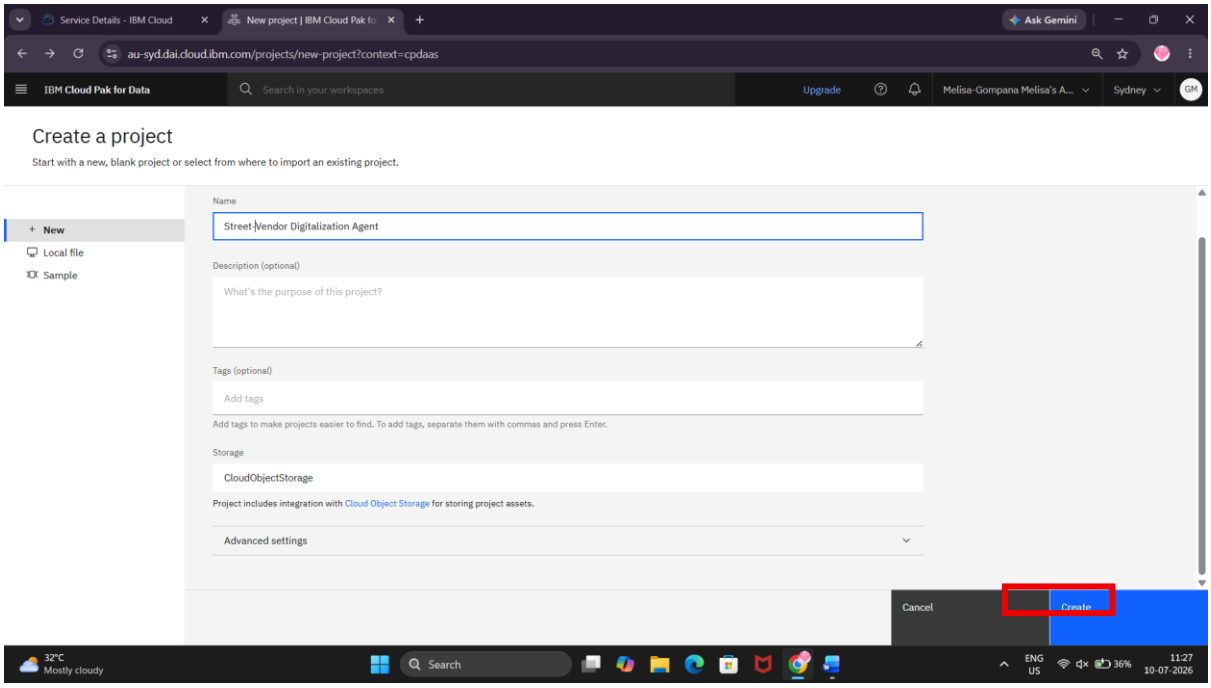
Step-6 select location and click on **create**



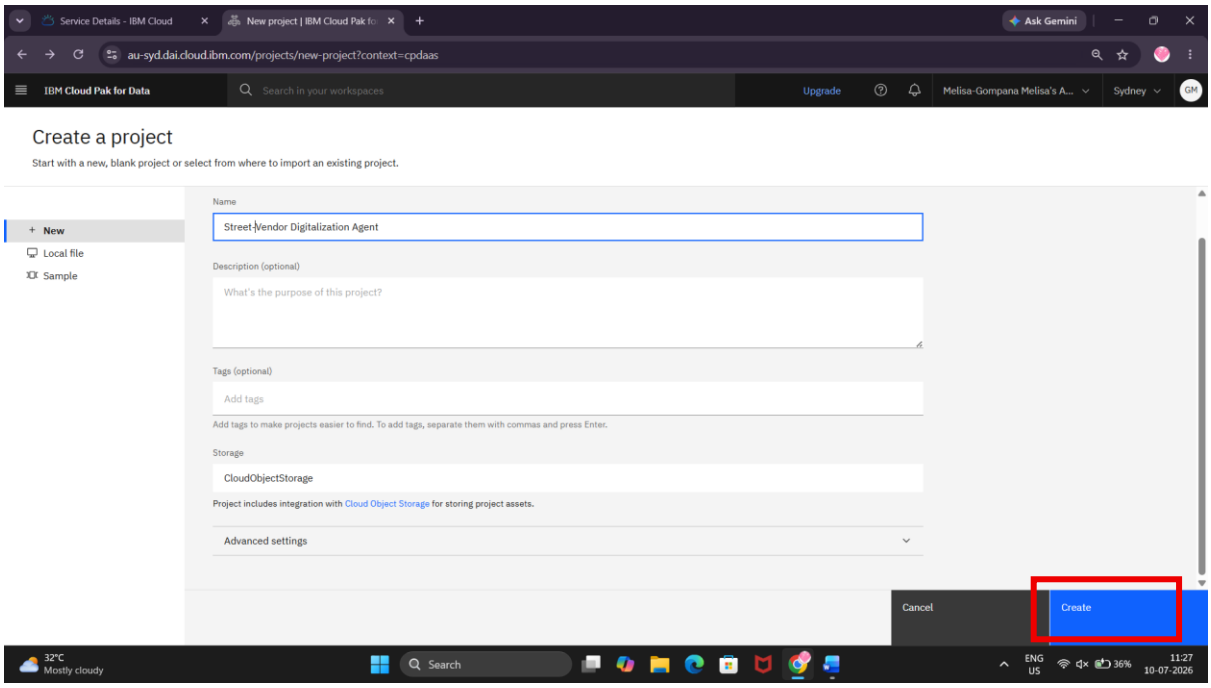
Step-7 Select **New project** and click on **next**.



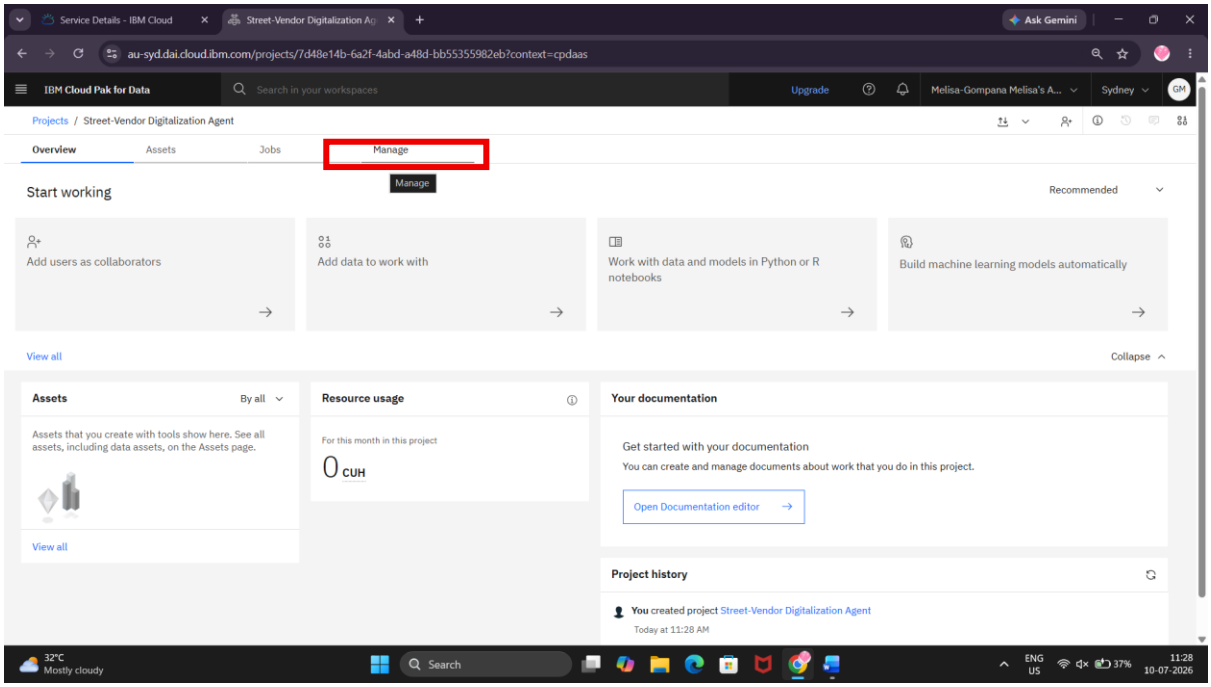
Step-8 give name to your project and click on **Create key**.



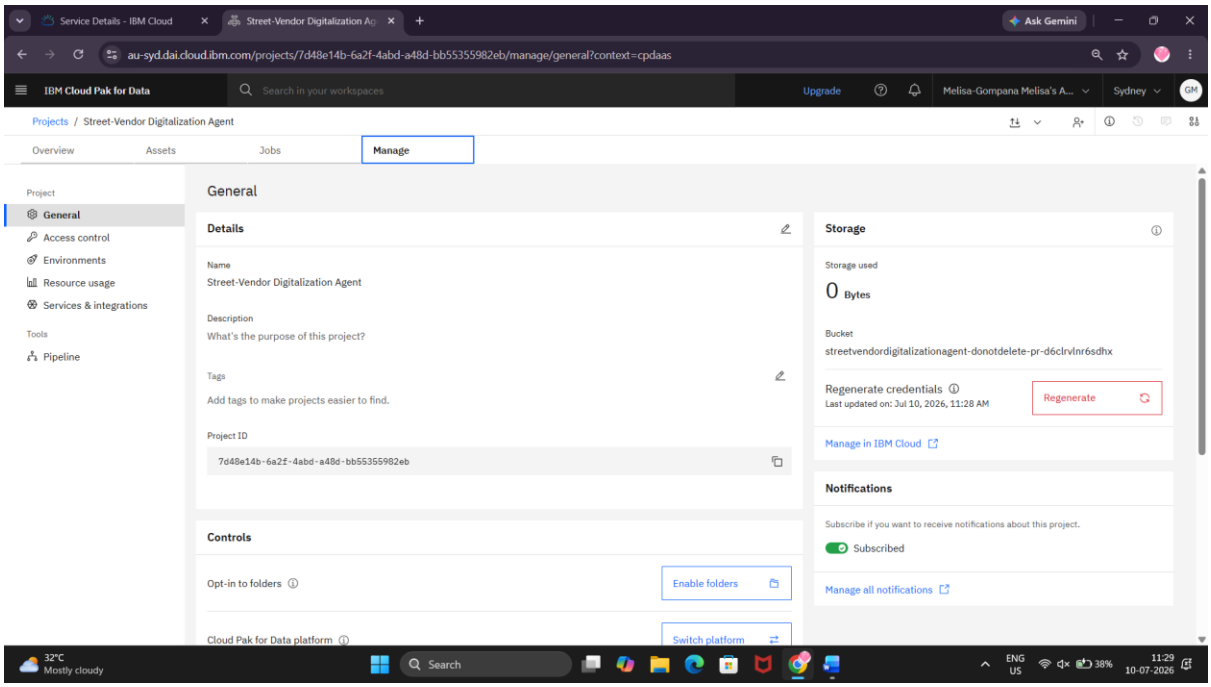
Step-9 click on **create**



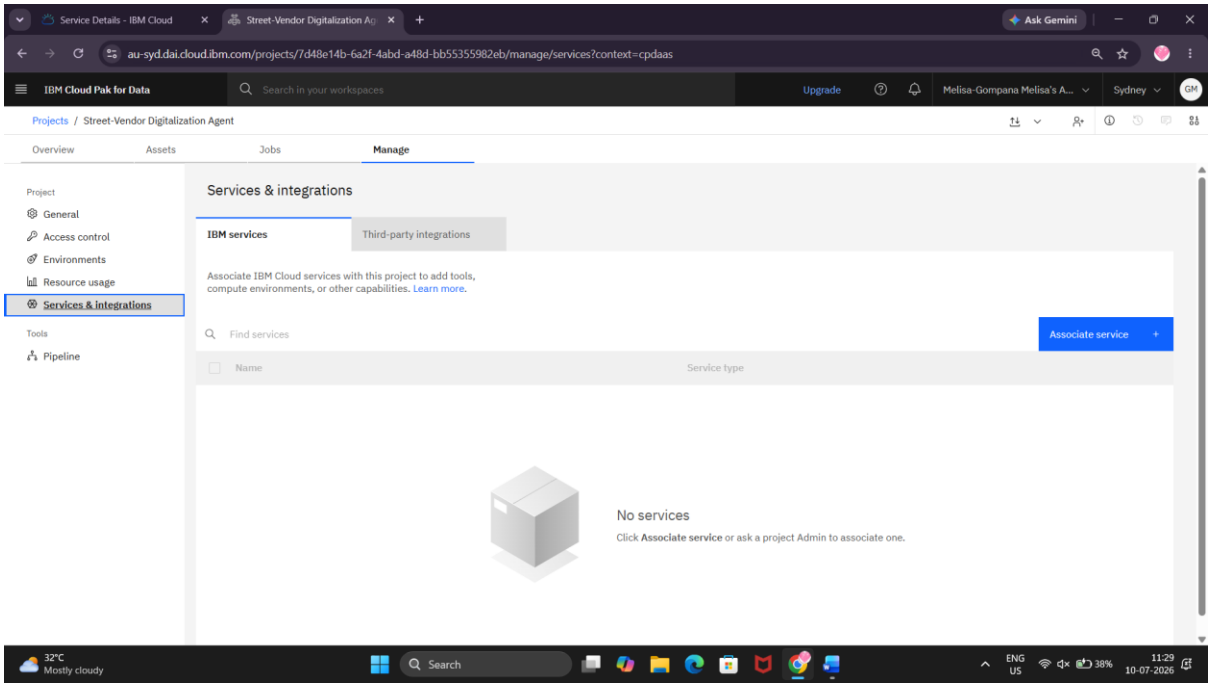
Step-10 go to **manage**



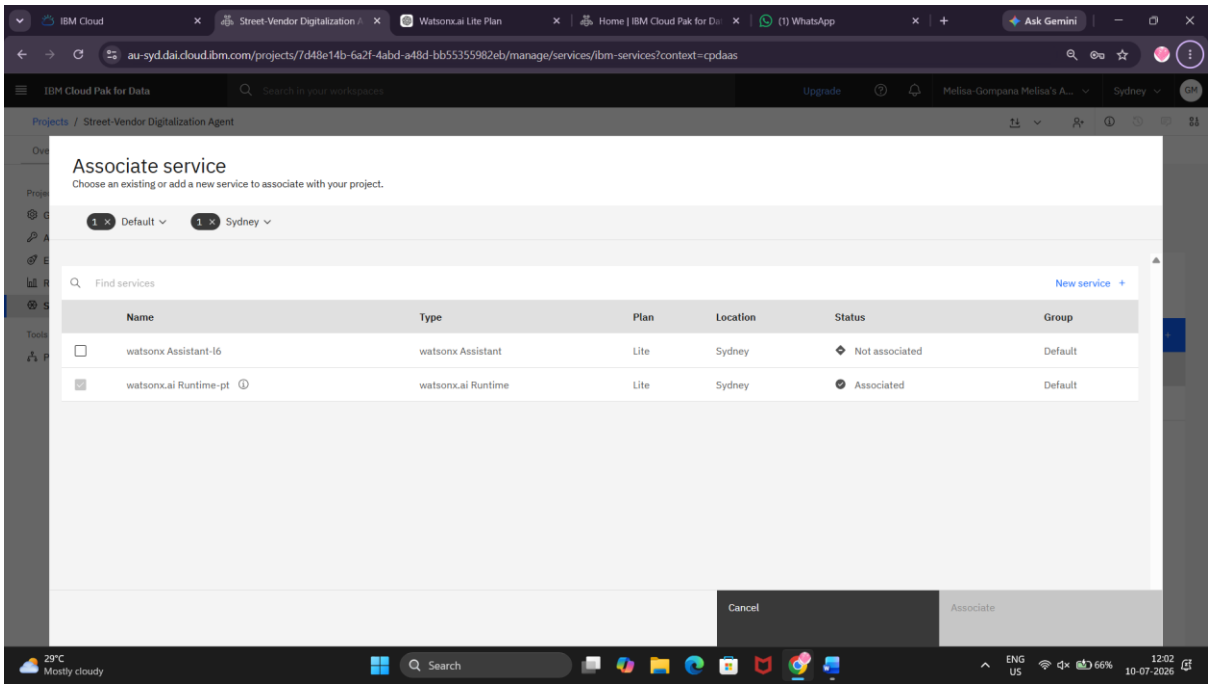
Step-11 click on **services & integration.**



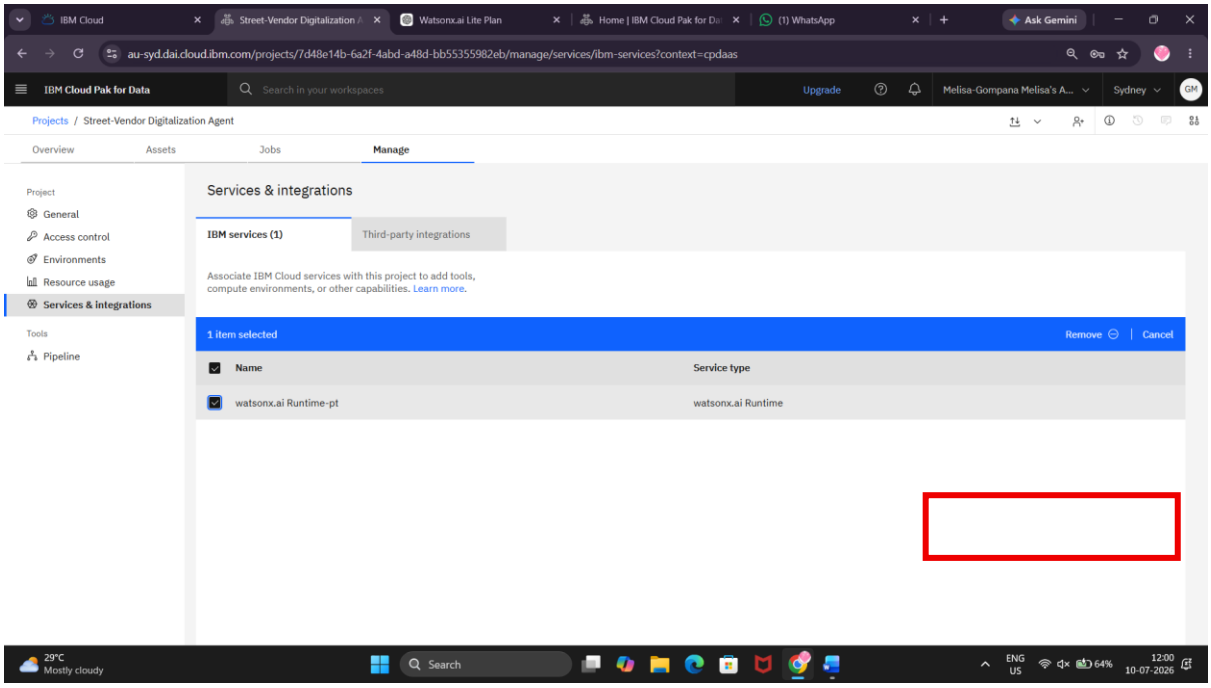
Step-12 click on **Associate Service**.



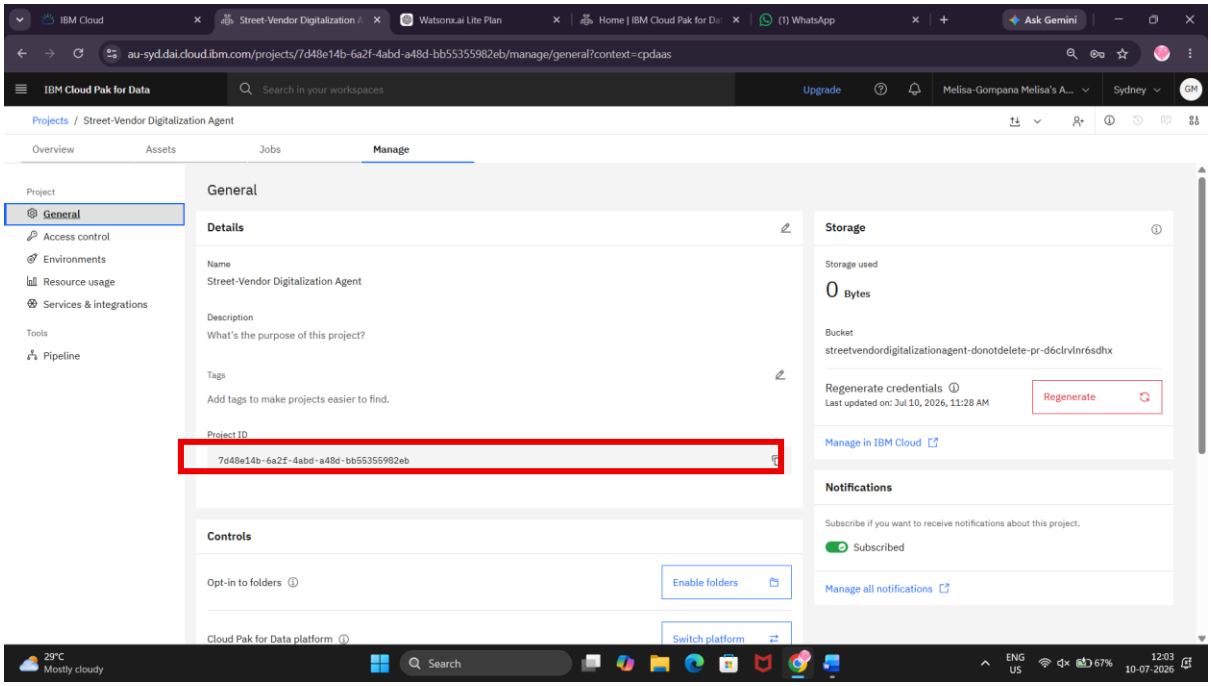
Step-13 mark the **checkbox**.



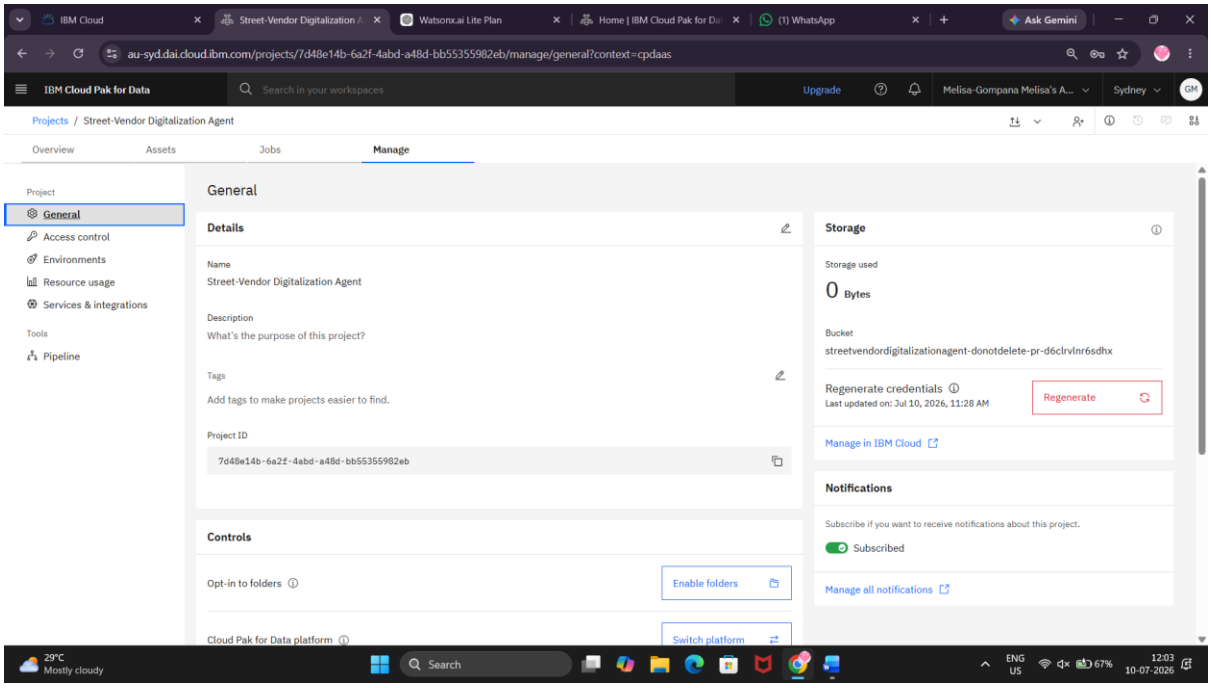
Step-14 click on **create**.



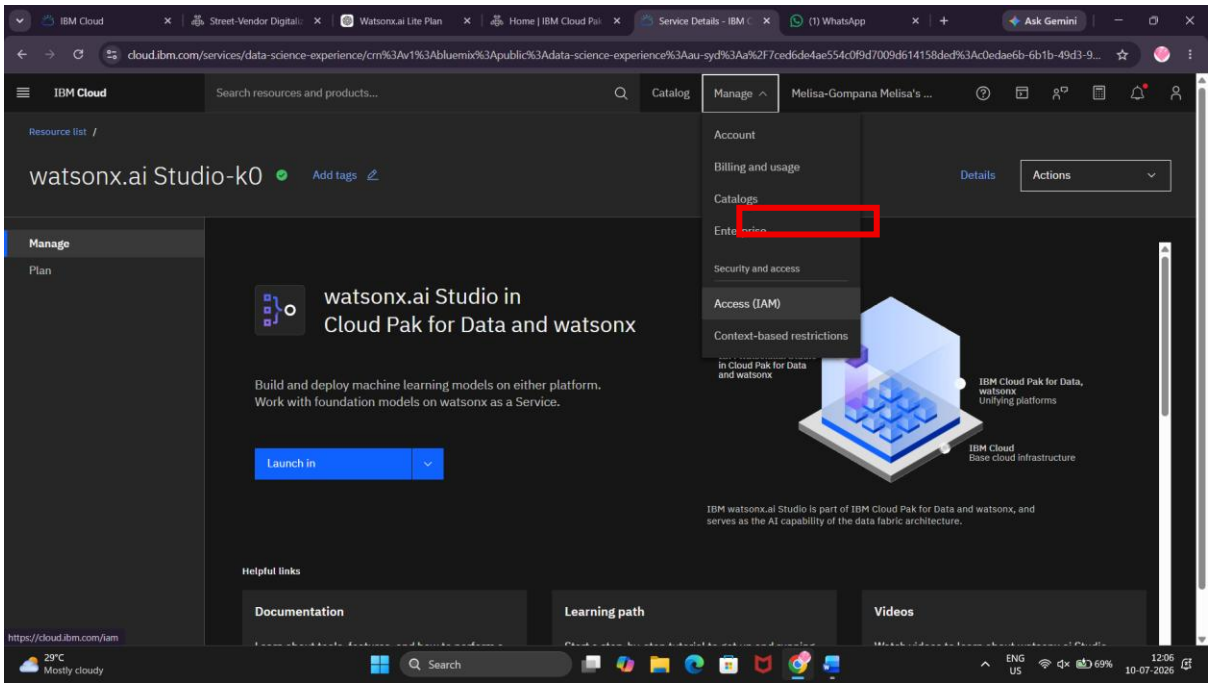
Step-15 Come to general and copy **project ID** and paste in your notepad.



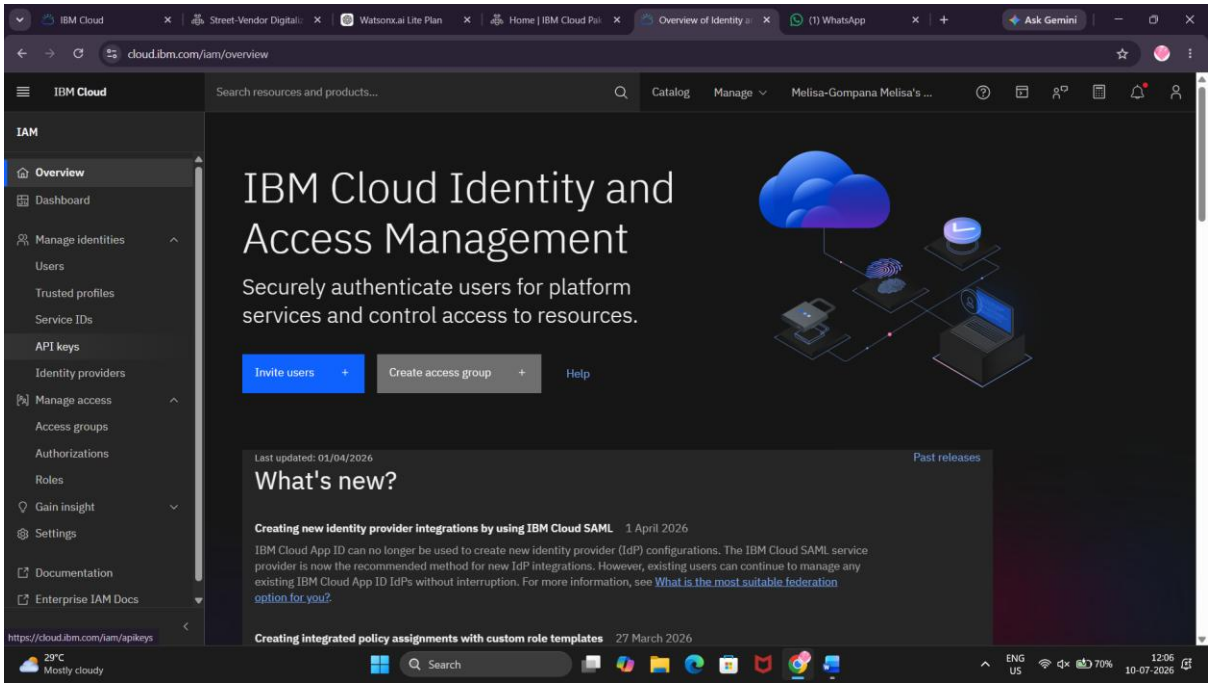
Step-16 Click on previous tab



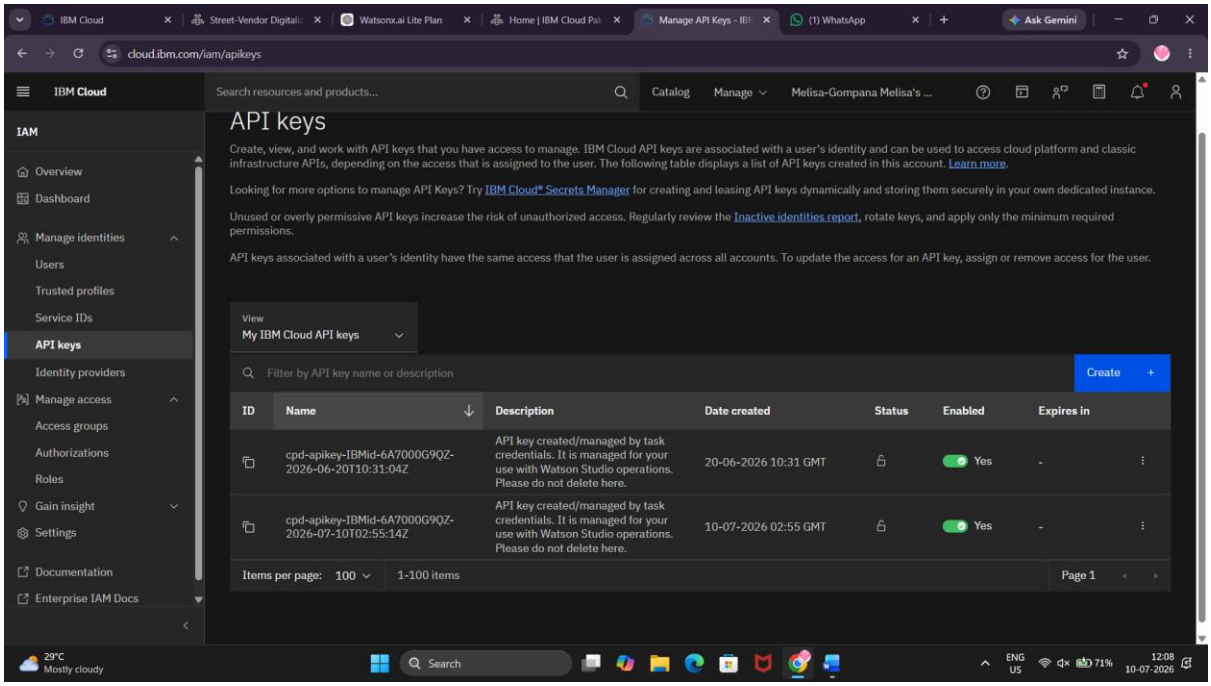
Step-17 click on **Access(IAM).**



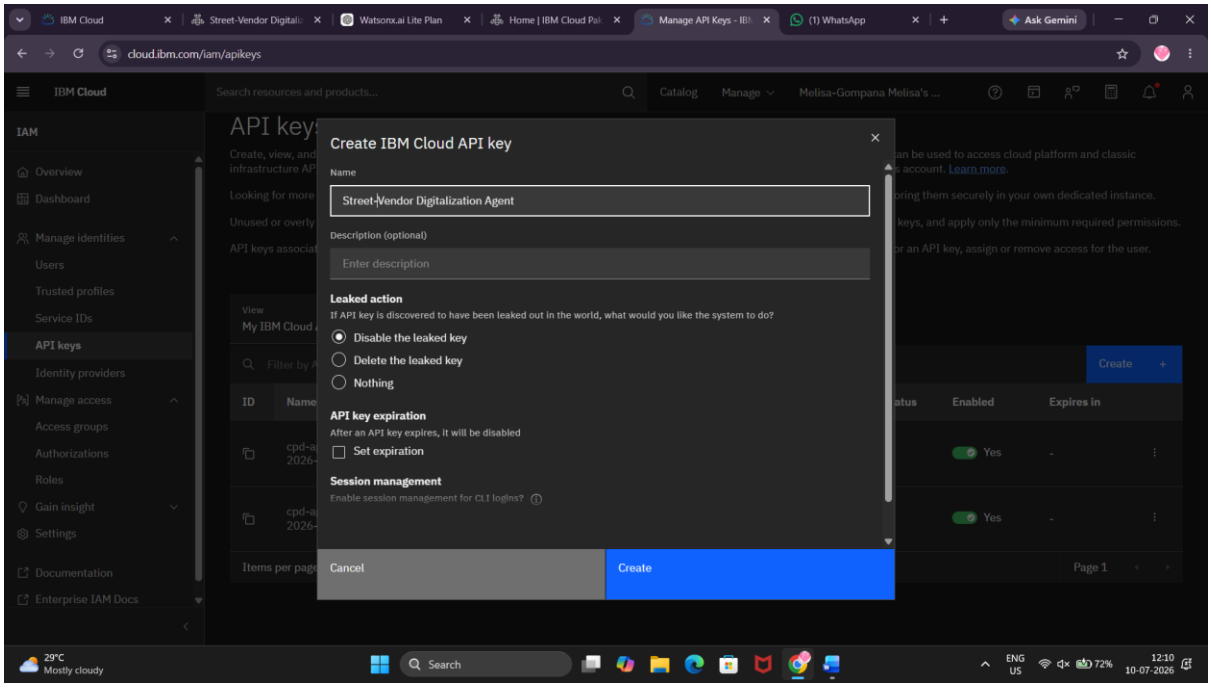
Step-18 click on **API keys**



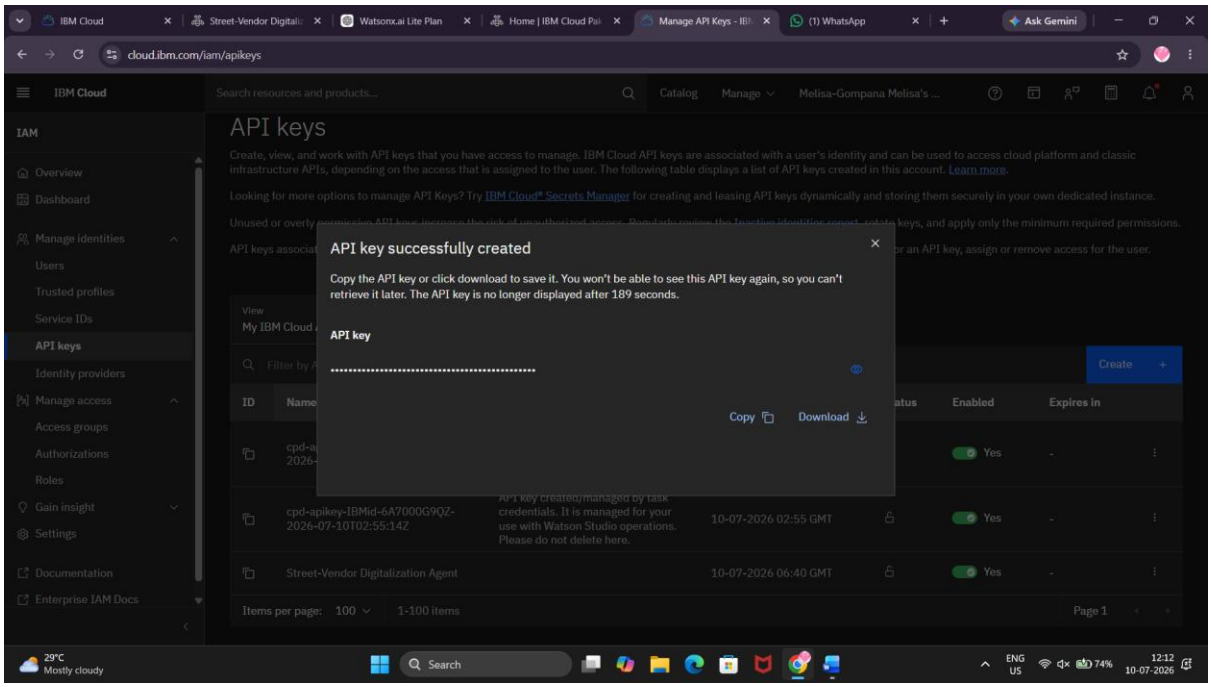
Step-19 click on **create** button



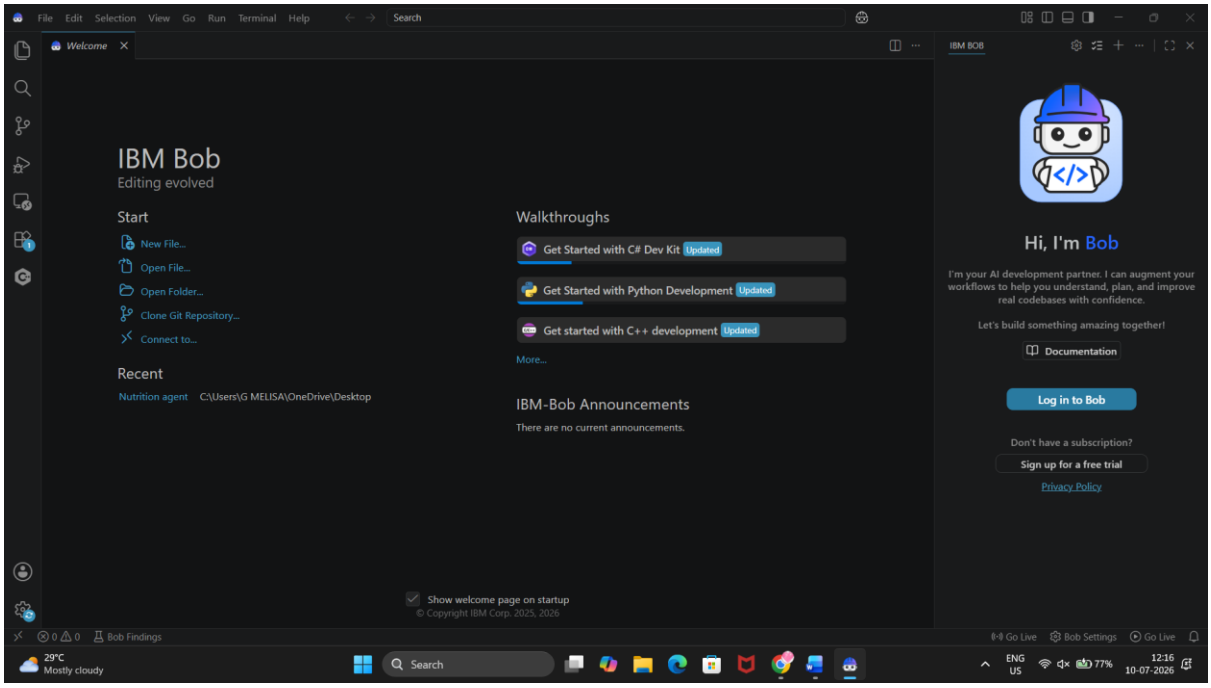
Step-20 give name to your api-key and click on **create**.



Step-21 Copy your **api-key** and paste in your notepad.



Step-22 Open IBM BOB.



Step-23 Give Prompt to your agent.

Prompt-

Create an AI-powered Street Vendor Digitalization Agent web application using Python Flask and IBM Watsonx.ai with IBM Granite models. Integrate the IBM Cloud API Key securely using a `.env` file and build a modern, responsive web application with an attractive chat interface and vendor dashboard.

The application should use Retrieval-Augmented Generation (RAG) to retrieve real-time information from government portals, MSME websites, fintech platforms, and commerce resources. Add an `AGENT_INSTRUCTIONS` section so the agent's behavior, response style, language, safety rules, and business recommendations can be easily customized.

The Street Vendor Digitalization Agent should provide the following features:

• Generate professional business profiles for street vendors and micro-entrepreneurs.

• Provide step-by-step UPI QR code setup guidance for digital payments.

• Recommend Google Business Profile registration and local online listing platforms.

• Generate Local SEO strategies to improve business visibility in nearby areas.

• Retrieve real-time MSME schemes, government policies, subsidies, and financial assistance programs using RAG.

• Suggest pricing strategies based on product category and local market conditions.

• Recommend digital marketing ideas, promotional offers, and customer engagement strategies.

• Generate promotional posters, business descriptions, and social media content.

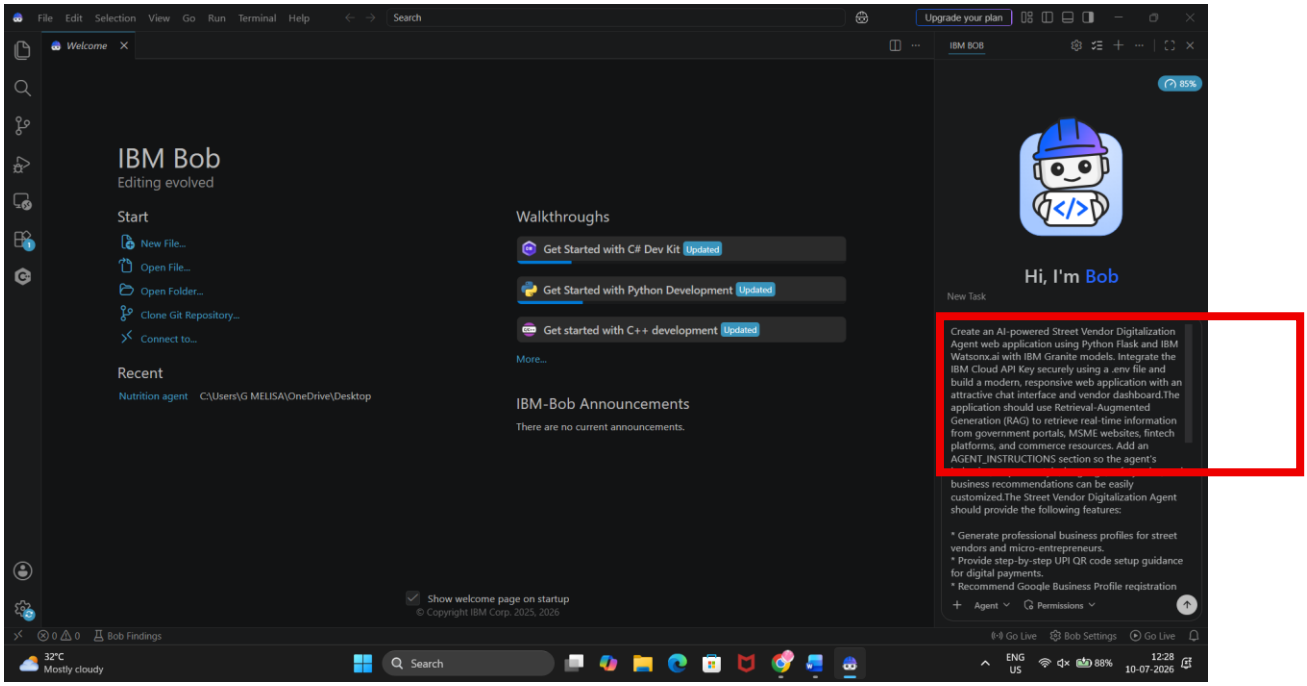
• Provide multilingual responses (English, Hindi, Telugu, Marathi, Tamil, etc.).

• Offer digital onboarding guidance for first-time vendors.

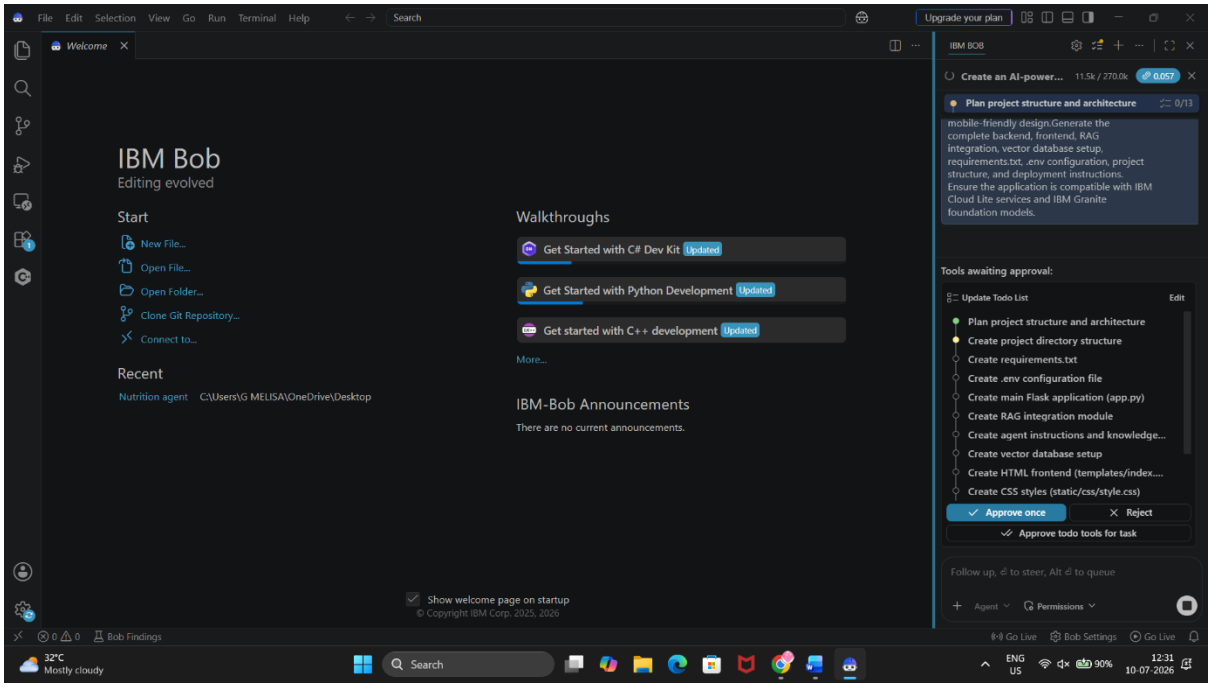
• Maintain conversation history and personalized vendor profiles.

• Include a responsive Bootstrap-based frontend with animations, dark mode, and mobile-friendly design.

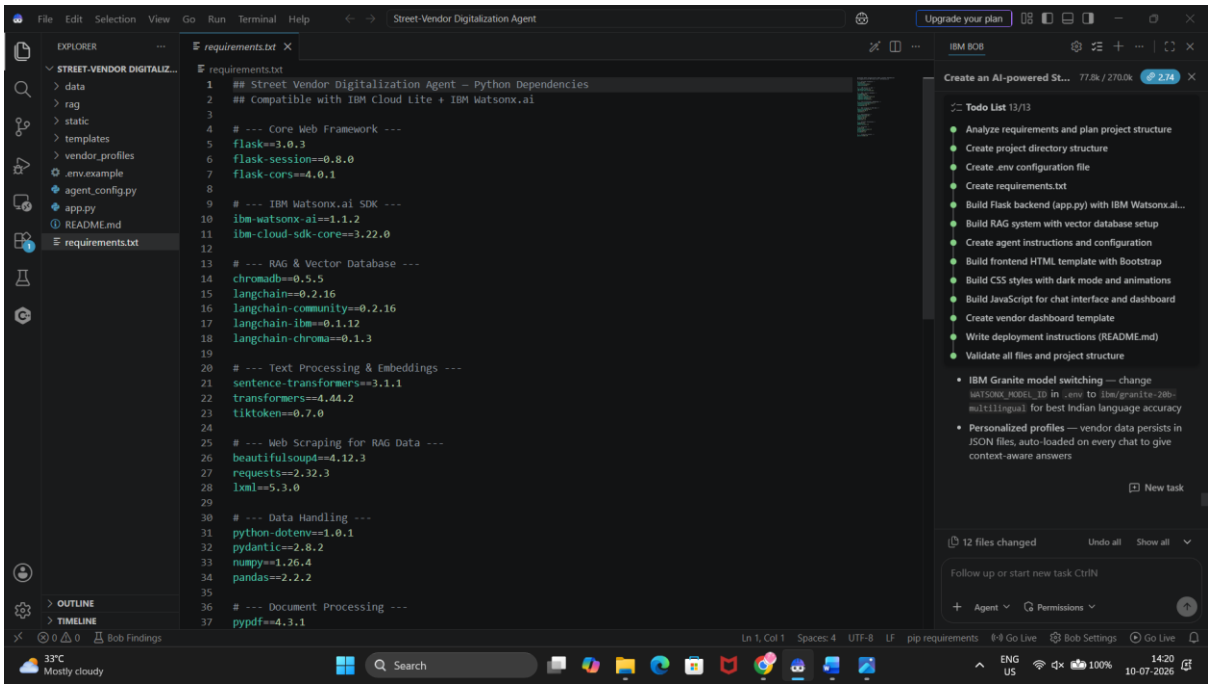
Generate the complete backend, frontend, RAG integration, vector database setup, requirements.txt, .env configuration, project structure, and deployment instructions. Ensure the application is compatible with IBM Cloud Lite services and IBM Granite foundation models.



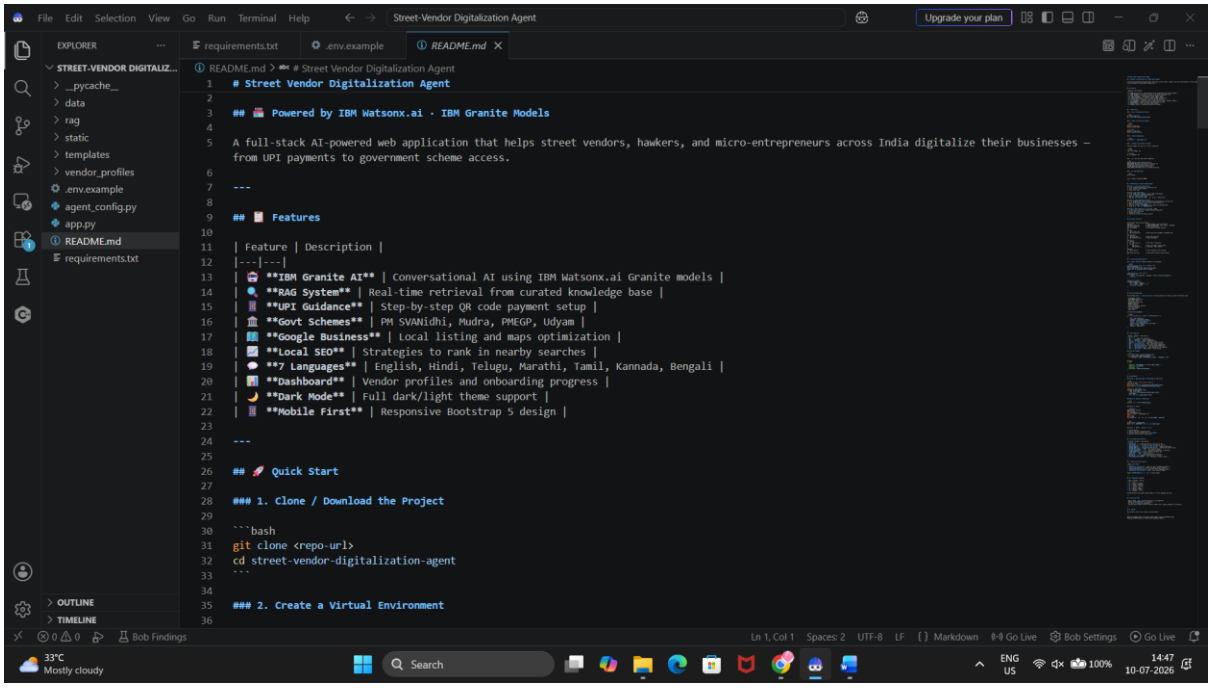
Step-24 Approve the plan created by BOB Agent



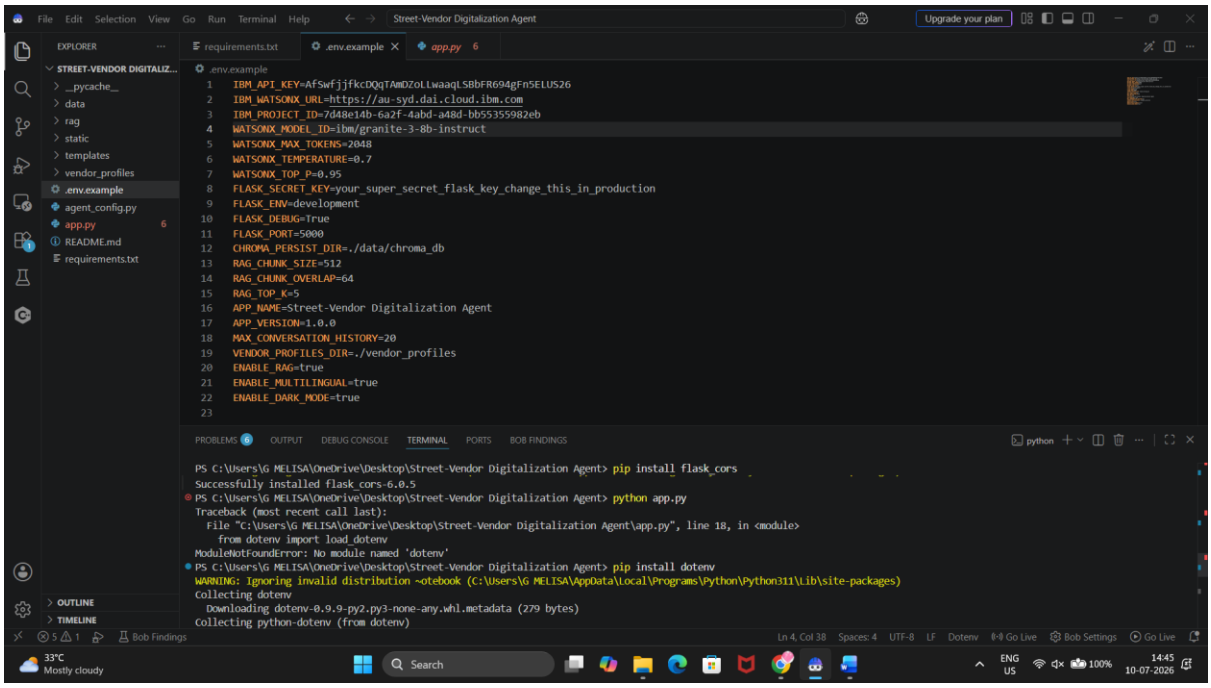
Step-25 Save the code give by BOB.



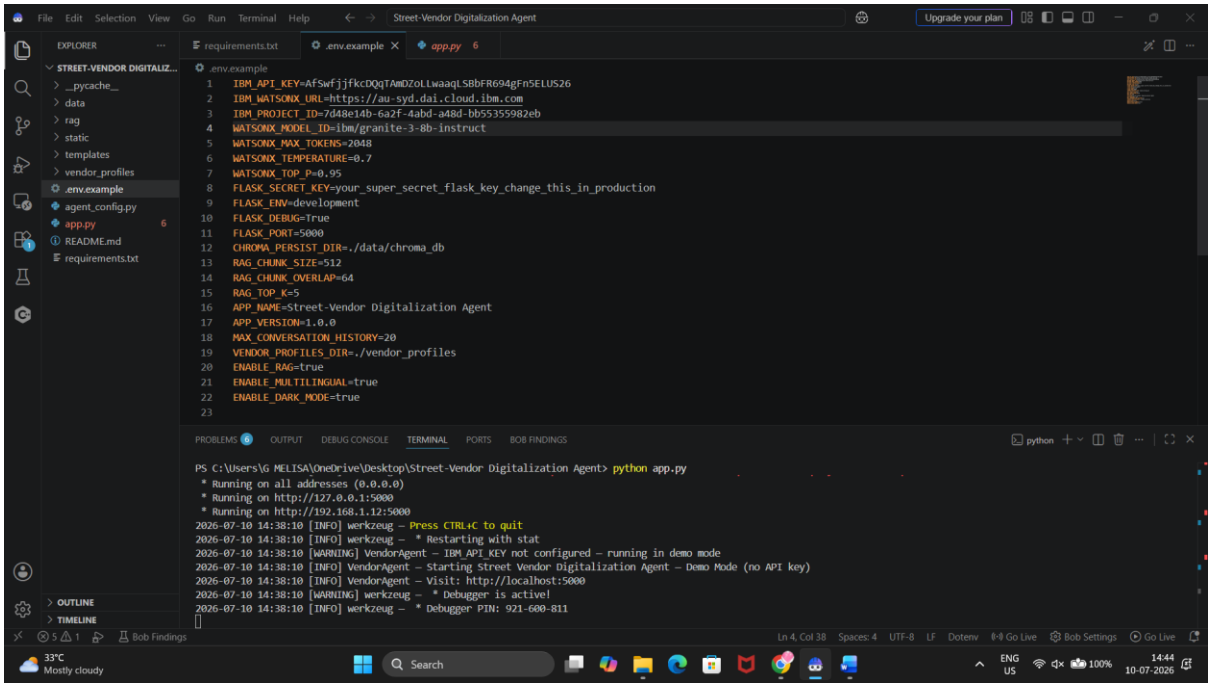
Step-26 It will create code form the scratch.



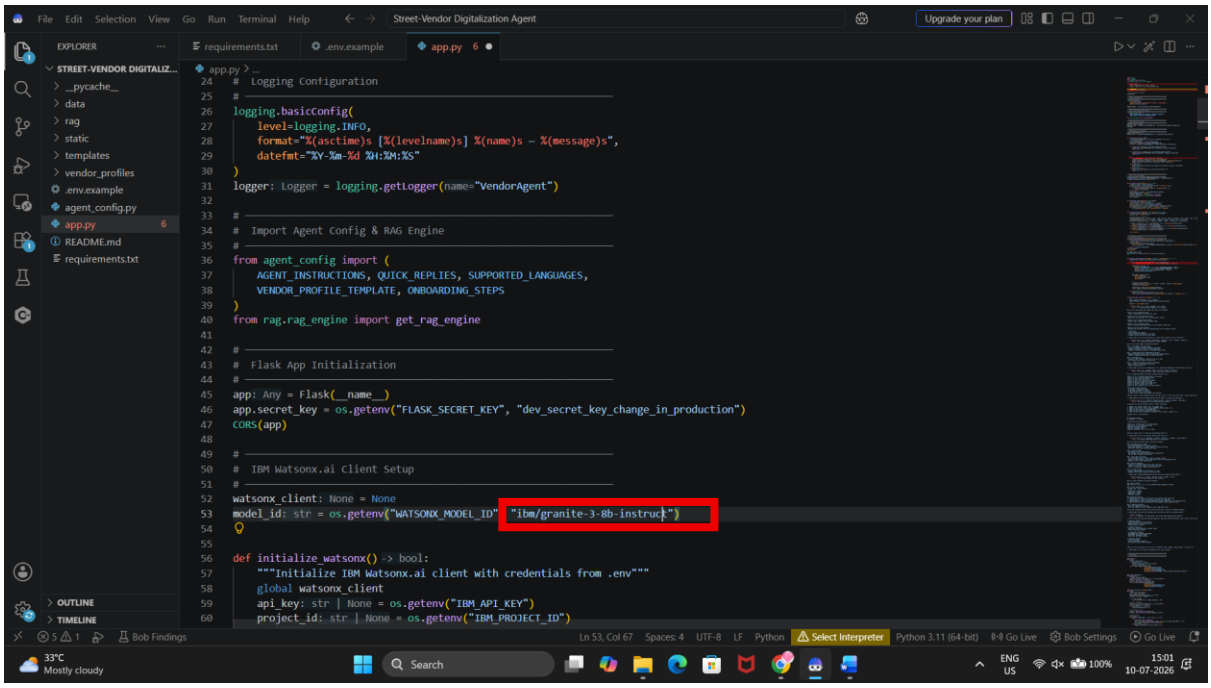
Step-27 once All file created approve for guideline and Now install All libraries in your system using UV.



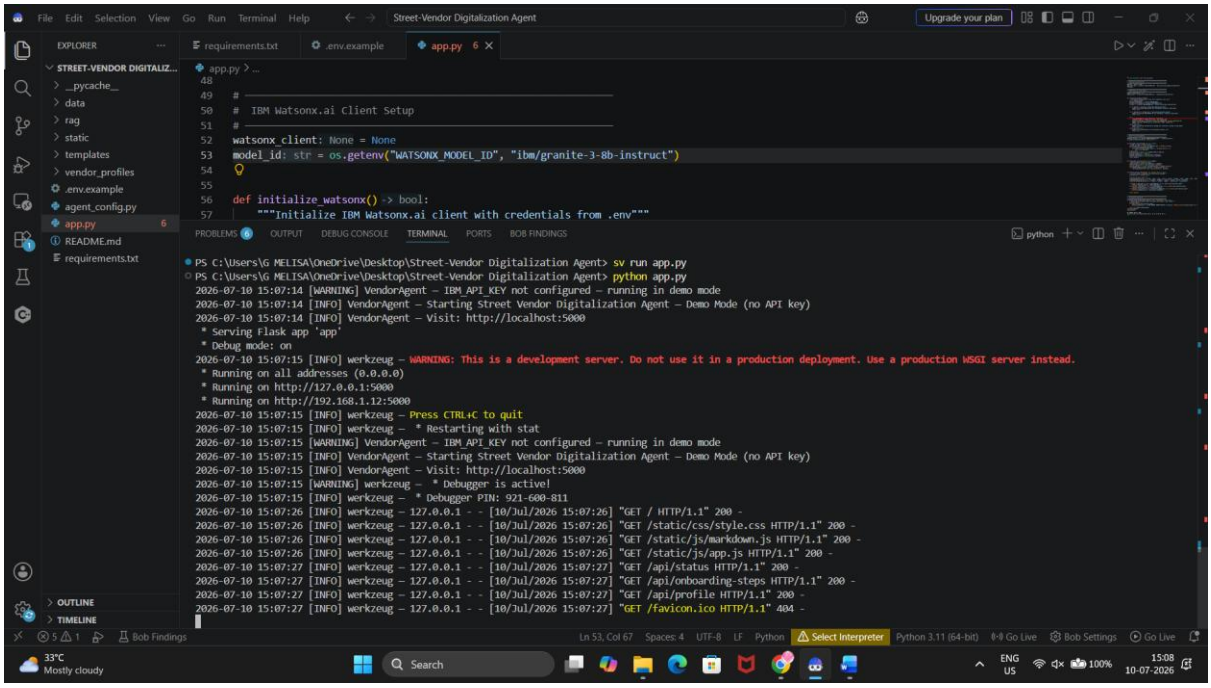
Step-28 Add **your API Key ,Project ID , IBM URL** and **Model Name in .ENV** file.



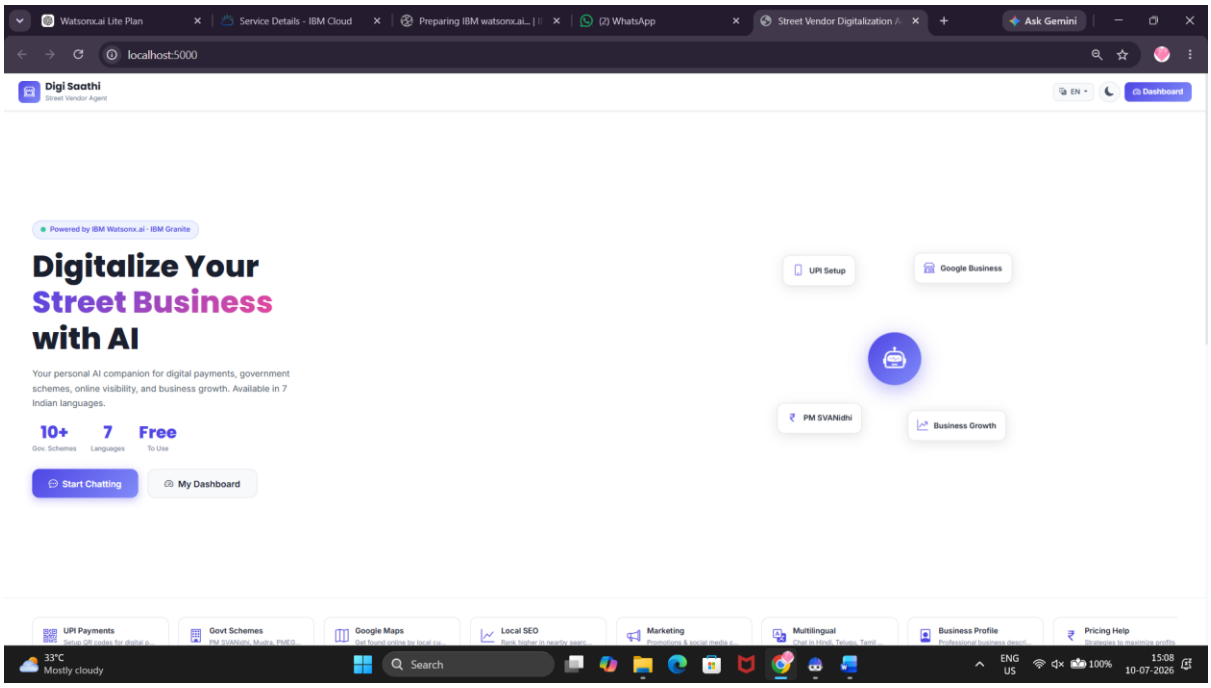
Step-29 Change **URL** Link and Model name From **App.py** .



Step-30 Run this Project Using Below Link And Click on Local Host URL
UV run app.py



Step-31 check Website create by BOB.

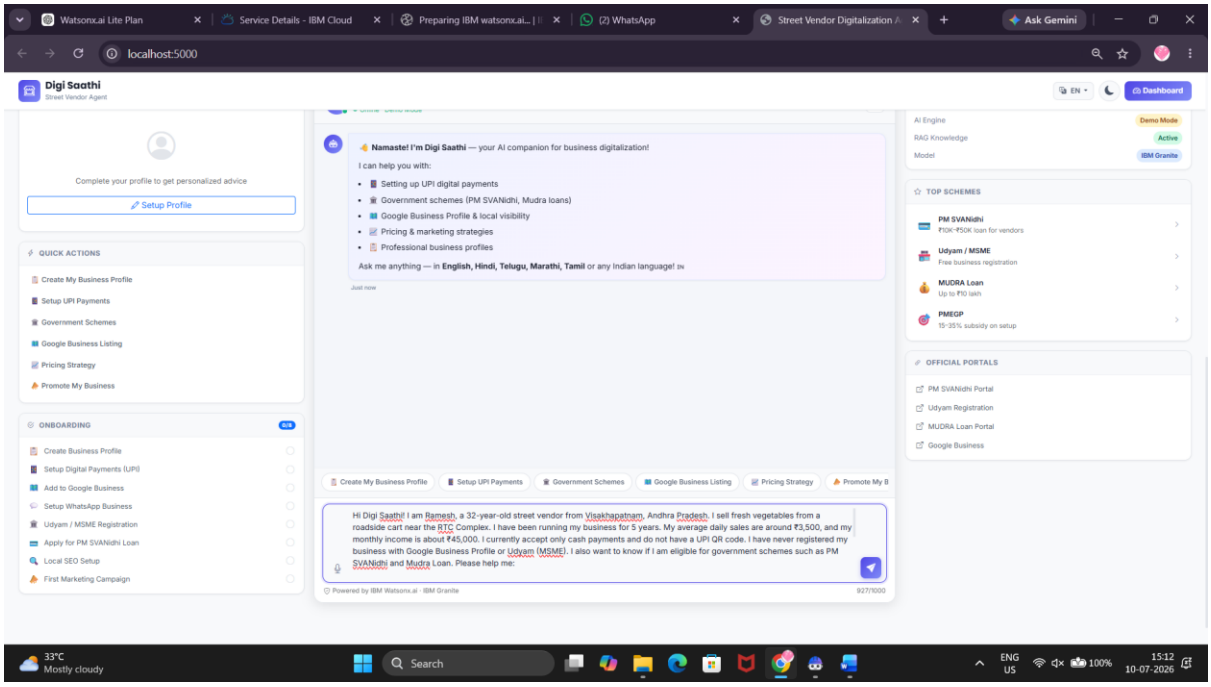


Step-32 Give Prompt and test your Agent

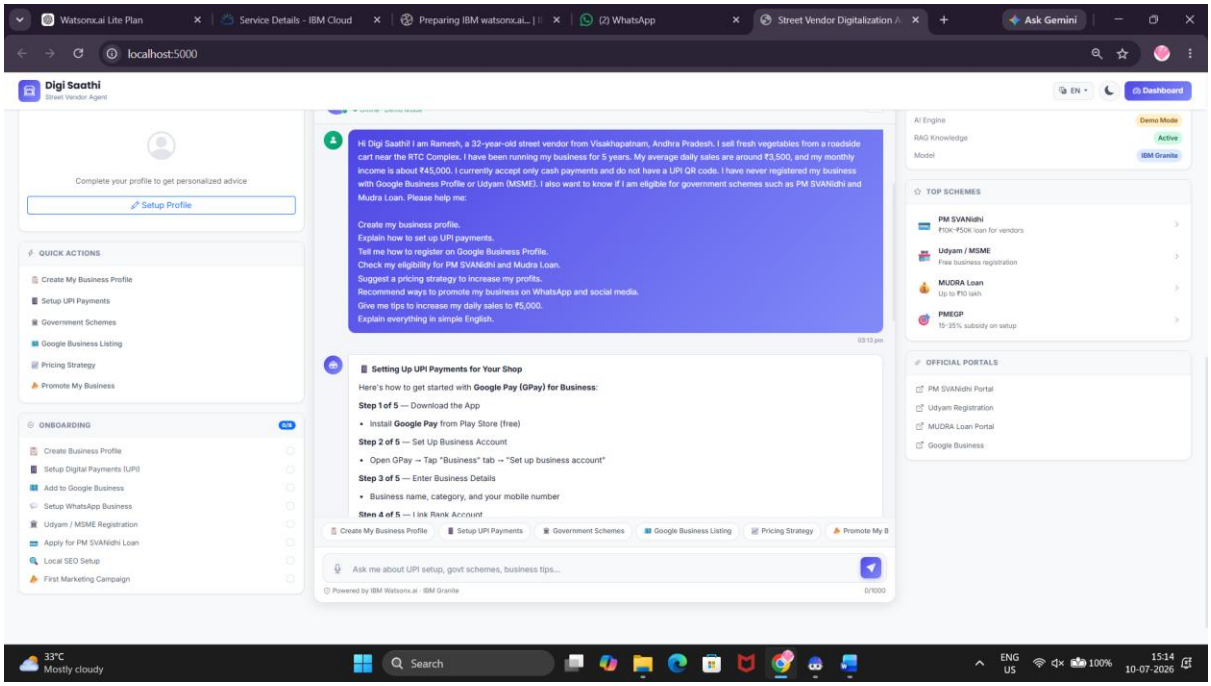
Hi Digi Saathi! I am Ramesh, a 32-year-old street vendor from Visakhapatnam, Andhra Pradesh. I sell fresh vegetables from a roadside cart near the RTC Complex. I have been running my business for 5 years. My average daily sales are around ₹3,500, and my monthly income is about ₹45,000. I currently accept only cash payments and do not have a UPI QR code. I have never registered my business with Google Business Profile or Udyam (MSME). I also want to know if I am eligible for government schemes such as PM SVANidhi and Mudra Loan. Please help me:

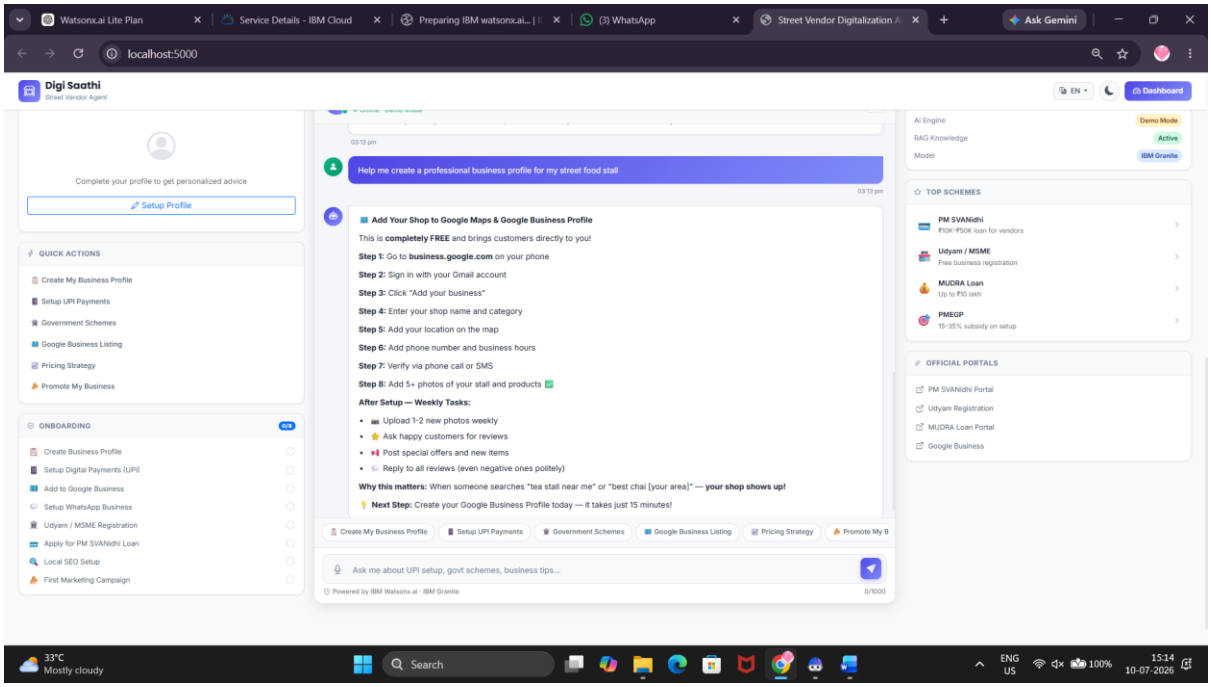
1. Create my business profile.
2. Explain how to set up UPI payments.
3. Tell me how to register on Google Business Profile.
4. Check my eligibility for PM SVANidhi and Mudra Loan.

5. Suggest a pricing strategy to increase my profits.
6. Recommend ways to promote my business on WhatsApp and social media.
7. Give me tips to increase my daily sales to ₹5,000.
8. Explain everything in simple English.



Step-33 Check the Output.




Watsonx.ai Lite Plan
Service Details - IBM Cloud
Preparing IBM watsonx.ai...
(3) WhatsApp
Street Vendor Digitalization
Ask Gemini
localhost:5000
Digi Saathi
Street Vendor Agent
EN
Dashboard
Complete your profile to get personalized advice
Setup Profile
QUICK ACTIONS
Create My Business Profile
Setup UPI Payments
Government Schemes
Google Business Listing
Pricing Strategy
Promote My Business
ONBOARDING
0/8
Create Business Profile
Setup Digital Payments (UPI)
Add to Google Business
Setup WhatsApp Business
Udyam / MSME Registration
Apply for PM SVANidhi Loan
Local SEO Setup
First Marketing Campaign
03:13 pm
Help me create a professional business profile for my street food stall
03:13 pm
Add Your Shop to Google Maps & Google Business Profile
This is completely FREE and brings customers directly to you!
Step 1: Go to business.google.com on your phone
Step 2: Sign in with your Gmail account
Step 3: Click "Add your business"
Step 4: Enter your shop name and category
Step 5: Add your location on the map
Step 6: Add phone number and business hours
Step 7: Verify via phone call or SMS
Step 8: Add 5+ photos of your stall and products
After Setup — Weekly Tasks:
Upload 1-2 new photos weekly
Ask happy customers for reviews
Post special offers and new items
Reply to all reviews (even negative ones politely)
Why this matters: When someone searches "tea stall near me" or "best chai [your area]" — your shop shows up!
Next Step: Create your Google Business Profile today — it takes just 15 minutes!
Create My Business Profile
Setup UPI Payments
Government Schemes
Google Business Listing
Pricing Strategy
Promote My B
Ask me about UPI setup, govt schemes, business tips...
Powered by IBM Watsonx.ai · IBM Granite
0/1000
AI Engine
Demo Mode
RAG Knowledge
Active
Model
IBM Granite
TOP SCHEMES
PM SVANidhi
₹10K-₹50K loan for vendors
Udyam / MSME
Free business registration
MUDRA Loan
Up to ₹10 lakh
PMEGP
15-35% subsidy on setup
OFFICIAL PORTALS
PM SVANidhi Portal
Udyam Registration
MUDRA Loan Portal
Google Business
33°C
Mostly cloudy
Search
ENG
US
100%
15:14
10-07-2026